Name: ____________________________ Partner: _______________________ Period: _________

Part A — Satellite Orbit

A satellite orbits Earth in a circular orbit of radius 8000 km, centered at Earth's center (the origin). It starts at (8000, 0) and travels counterclockwise, completing one full orbit in 2 hours. Using t in hours, the angle after t hours is πt radians.

Tier R — Remediate

A1. Write parametric equations $x(t)$ and $y(t)$ for the satellite's position at time t hours.

$x(t) = 8000\cos(\pi t)$ $\quad$ $y(t) = 8000\sin(\pi t)$

A2. Complete the table.

t (hours)	$x(t)$ (km)	$y(t)$ (km)	Position $(x,\ y)$
0	8000	0	$(8000,\ 0)$
0.5	0	8000	$(0,\ 8000)$
1	-8000	0	$(-8000,\ 0)$
1.5	0	-8000	$(0,\ -8000)$
2	8000	0	$(8000,\ 0)$

Tier A — Approaching Proficiency

A3. At what time t is the satellite directly above the origin on the north side ($x = 0$, $y > 0$)?

Set $x(t) = 8000\cos(\pi t) = 0$: $\cos(\pi t) = 0$, **so** $\pi t = \pi/2$, **giving** $t = 1/2$ **hour.** At $t = 0.5$: $y(0.5) = 8000\sin(\pi/2) = 8000$ **km** > 0. **The satellite is directly north of the origin at** $t = 0.5$ **hours.**

A4. Clockwise orbit: replace t with $-t$:
$x(t) = 8000\cos(-\pi t) = 8000\cos(\pi t)$ (same)
$y(t) = 8000\sin(-\pi t) = -8000\sin(\pi t)$

Teacher Note

A3 is a good cold-call opportunity. A4 reinforces the key even/odd identity: $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.

Part B — Rover Path

A rover travels in a straight line from base camp at $(-3,\ 2)$ to a sample site at $(5,\ -2)$.

Tier E — Extension

B1. Parametric equations for $0 \le t \le 1$:
$x(t) = -3 + (5 - (-3))t = -3 + 8t$
$y(t) = 2 + (-2 - 2)t = 2 - 4t$

B2. Positions:
At $t = 0.25$: $x = -3 + 8(0.25) = -1$, $y = 2 - 4(0.25) = 1$. Position: $(-1,\ 1)$.
At $t = 0.75$: $x = -3 + 8(0.75) = 3$, $y = 2 - 4(0.75) = -1$. Position: $(3,\ -1)$.

B3. Rover crosses x-axis when $y(t) = 0$:
$2 - 4t = 0 \ \Rightarrow\ t = 1/2$.
At $t = 1/2$: $x(1/2) = -3 + 8(1/2) = 1$.
The rover crosses the x-axis at the point $(1,\ 0)$, when $t = 0.5$.

Teacher Note

B2–B3 preview the idea that the parameter t can be solved for algebraically (EK 4.4.A.3). Connect to Lesson 4.1's skill of solving for t to find when a curve hits a target coordinate.